

TrayGuard — Surgical Tray Training

EC ENGR 180DW — Final Design Review

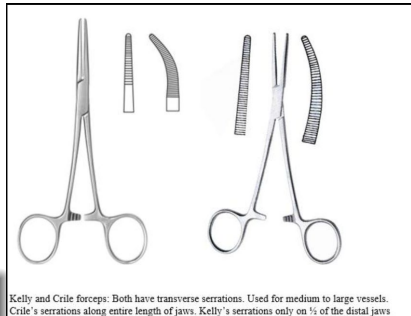
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June 2026

The Problem

- **9 defects per 100** surgical cases
- **55%** occur during tray assembly
- **10 min** average delay per affected case
- **\$6.75–9.42M** estimated annual cost per hospital
- **44%** of packaging errors: wrong specification

“The total count was correct. The specific instrument was wrong.”

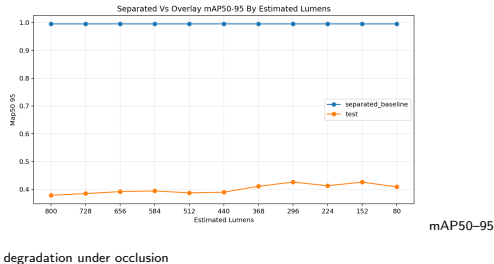


Kelly vs Crile —

same family, different serration pattern

Why Cameras Failed

- Seven staged experiments: brightness, background, layout occlusion
- **Every real condition** degraded detection
- Occlusion: **mAP50–95 dropped to 0.389**
- The model learned **positions, not shapes**
- Site-specific data needed per hospital — not feasible for one project

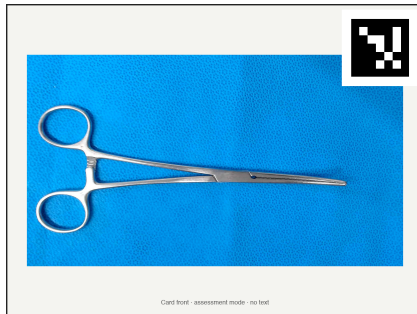


The data told us no.

TrayGuard

- Training platform: **laptop + printer + cards**
- Camera scans which cards are on the table
- Scoring engine classifies every selection into **6 error categories**
- Confidence capture: separates confidently wrong from fragile knowledge
- No special equipment. No hospital deployment required.

The real bottleneck is training the people who build trays.



Clamp Kelly 8"

Family: Clamping/occluding · Catalog: CLAMP KELLY 8"

Key identification features

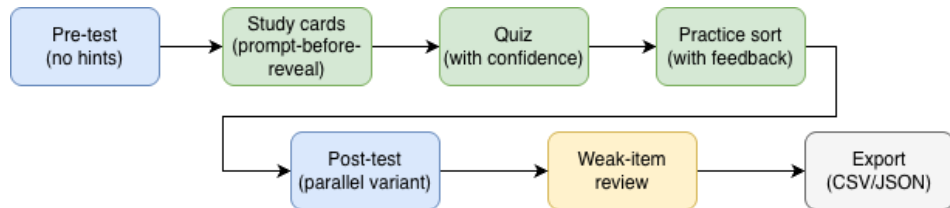
- Curved jaws with ring handles
- Transverse serrations on distal half of jaws
- 8-inch overall length

Common lookalikes

- Crile clamp — serrations run full jaw (Kelly: distal half only)
- Kocher clamp — single tooth at tip (Kelly: no tooth)
- Kelly 6" / Kelly 12" — same jaw shape, different length

Also known as: JARIT_106-191

How It Works



Pre / post use parallel variants. Study cards are retrieval practice. Practice gives error-specific feedback.

Pre / Post Test

Parallel variants
Different photo views
No hints
Baseline + measured gain

Study + Quiz

Retrieval-first cards
Prompt-before-reveal
Immediate feedback
Confidence capture

Practice Sort

Physical cards on table
Scored against count sheet
6 error categories
Weak-item review

Prototype built. Study protocol designed.

**Looking for a hospital partner
to pilot the first real tray module.**

Open-source. Documented design handoff.

Contact the TrayGuard team through UCLA EC ENGR 180DW.

Direct material cost: **\$53**. Requires: printer + existing laptop or tablet.

Evidence: Error by the Numbers

Alfred et al. (2021)

3,900 defects
41,799 cases

9 per 100
55% at assembly

Nichol et al. (2024)

236 errors
147 cases

10 min avg delay
\$6.75–9.42M / yr

Zhu et al. (2019)

398 errors
33,839 packages

44% wrong spec
Largest error category

Three different methods converge on the same finding:
tray errors are recurring, measured, operational failures.

Three methods. One finding.

The Lookalike Problem: Kelly vs Crile



Kelly and Crile forceps: Both have transverse serrations. Used for medium to large vessels.
Crile's serrations along entire length of jaws. Kelly's serrations only on ½ of the distal jaws

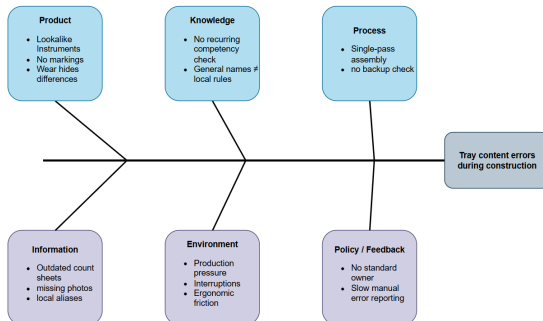
Kelly (left) vs Crile

(right) — serration pattern is the discriminating feature

- Same instrument family: both hemostatic clamps
- **Crile:** serrations cover the full jaw
- **Kelly:** serrations only cover distal half
- Sizes vary: 6", 7", 8"
- 44% of packaging errors are wrong specification

A few millimeters of serration — invisible at a glance.

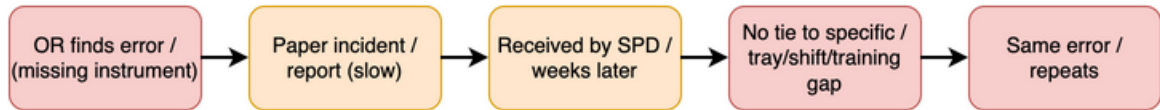
Root Cause Analysis



Six categories: instrument design, knowledge, process, information, environment, policy.
Three structural drivers: SPD as cost center, no licensure, no OR→SPD information system.

Six causes. Three structural drivers.

The Broken Feedback Loop



Evidence: 7 Staged CV Experiments

Experiment	Condition	Impact
Brightness (train dark, test bright)	80 $\rightarrow$ 800 lumens	mAP degrades with gap
Brightness (train bright, test dark)	800 $\rightarrow$ 80 lumens	Same pattern, symmetric
Background reflectivity	Matte vs reflective	Matte best, reflective worst
Layout: separated $\rightarrow$ separated	Clean lab layout	mAP50 = 0.990
Layout: separated $\rightarrow$ overlay	Instruments touching	mAP50-95 = 0.389
Real instruments (kitchen proxy)	6 classes, fixed positions	Per-class AP 0.000-0.862
Real instruments (position check)	Split-1 vs Split-2	Classes survive different splits

Every real condition degraded detection.

Evidence: Occlusion — Before and After



Separated — what

you train on



Overlay — what a

real tray looks like

You'd have to rearrange instruments to scan them.

Evidence: Position vs Shape Learning

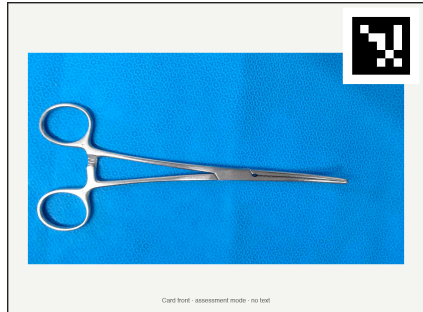
- Per-class AP ranged from **0.000 to 0.862** depending on data split
- Different instrument classes survived in different splits
- Only the **scalpel** transferred — the most visually distinct
- Each image had one of each class at a **fixed position** per session
- Model learned **which class at which location**, not visual features
- When position changed (different tray), the model failed

The model memorized positions, not shapes.

The Pivot: TrayGuard

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Design: Error Categories

- **Correct** — item present, correct count
- **Missing** — required item not selected
- **Extra** — selected item not required
- **Wrong** — wrong instrument family
- **Misidentified** — lookalike substitution
- **Wrong Count** — correct instrument, wrong quantity

Published evidence

- **Alfred et al.** — missing, extra, wrong
- **Nichol et al.** — wrong, delays
- **Zhu et al.** — 44% misidentified

Each category maps to published literature.
When a learner improves, we know *which type* of error they reduced.

Same categories hospitals already track.

Design: Confidence Capture

- Confidence rated per item: **1–5 scale**
- Plus a **“Not sure”** flag
- **High-confidence error**: confidently wrong
 - Hardest to correct — learner doesn't know they need help
- **Low-confidence correct**: fragile knowledge
 - May not survive a week without review
- Both are exported in CSV/JSON

Confidence: How sure are you?

Rate your confidence for each tray selection:

1	2	3	4	5
Guessing	Unsure	Neutral	Fairly sure	Certain

Not Sure

Confidently wrong is not the same as fragile knowledge.

Scope Boundaries

What TrayGuard Does

- Local tray familiarization
- Lookalike discrimination practice
- Simulated tray sort with error-category feedback
- Pre/post measurement with CSV/JSON export

What It Does NOT Do

- Replace CRCST certification
- Inspect instrument damage or wear
- Fix staffing shortages or wages
- Close the OR→SPD feedback loop

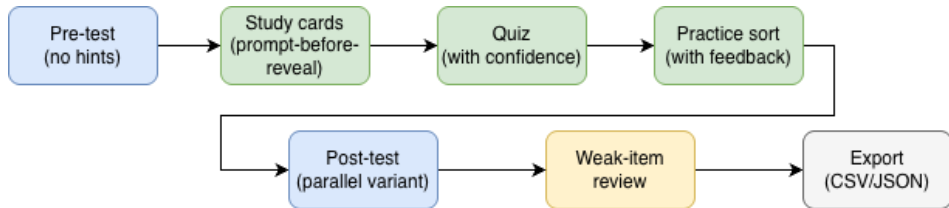
Claim bounded to: improvement on a simulated task in one session.

Alternatives Considered

Approach	Why not buildable by a student team
Automated tray checking (CV)	Per-hospital datasets, cross-manufacturer validation, regulatory review
Physical error-proofing (foam)	Requires instrument inventory, manufacturer relationships, per-tray tooling
Workflow redesign (dual check)	Requires policy authority, institutional buy-in, staffing changes
Decision-support kiosks	Requires deployment access, workflow integration, multi-stakeholder approval
Tray rationalization	Requires hospital utilization data, surgeon preferences, multi-department coordination
Competency tracking systems	Requires IT integration, HIPAA compliance, ongoing institutional commitment
Training	Buildable with cards + laptop. Validatable with pre/post study. Creates durable local content any future solution needs.

Training was the only buildable path.

Design: How It Works



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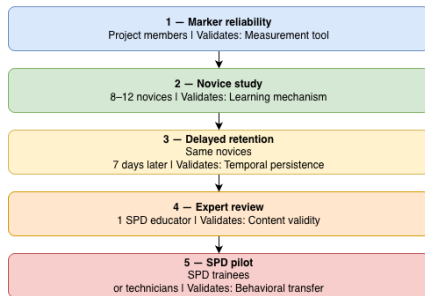
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Evaluation: The Study Ladder

Target: Kirkpatrick Level 2 (Learning). Levels 3–4 require future validation.



- Study 0** Marker reliability — do the cards scan?
- Study 1** Novice baseline — within-subject pre/post, 8–12 participants
- Study 2** 7-day retention — does improvement persist?
- Study 3** Expert review — SPD educator validates content
- Study 4** SPD workplace pilot — test with real technicians

Each step validates the next.

Evaluation: Falsification Criteria

For the 7-day retention check (Study 2):

- ① **Delayed accuracy retains at least half** of the immediate gain
- ② **At least 5 percentage points above baseline**
- ③ **At least 70% of participants return** for the follow-up

If any criterion fails: the claim reverts to **“immediate learning only.”**

We specify upfront what would falsify our claim.

What We Accomplished

- **7 staged CV experiments** with documented negative result
- **Root cause analysis** across 6 categories + 3 structural drivers
- **Full system specification:** 7-step learning loop, scoring engine, confidence capture, CSV/JSON export
- **Study protocol** with explicit falsification criteria
- **Prototype web app** (FastAPI + vanilla JS) implementing the full learner loop
- **Durable tray module schema** — JSON files capturing local instrument knowledge

A rigorous no on CV is a real finding.

Prototype built. Study protocol designed.

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to pilot the first real tray module.**

Open-source. Documented design handoff.

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Direct material cost: **\$53**. Requires: printer + existing laptop or tablet.

Q&A: How Is This Different from Flashcards?

Feature	Flashcards	TrayGuard
Assessment	Recall only	Physical tray sort against count sheet
Feedback	Binary right/wrong	6 error categories mapped to hospital tracking
Context	Isolated instrument name	Distractors, lookalikes, required quantities
Confidence	None	1–5 scale + “Not sure” flag
Output	None	CSV/JSON export with hashed learner ID

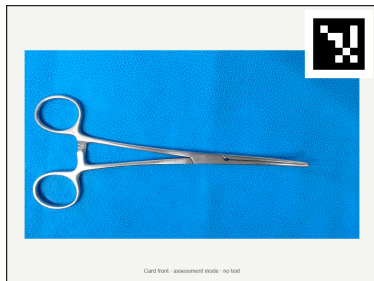
The applied task, not just memorization.

Q&A: Why AprilTags?

- **Faster detection** than QR codes — fewer bits to decode
- **Tolerates glare and partial occlusion** through built-in error correction
- **Stable 6-DOF pose estimation** handles camera angle variation
- **Manual fallback:** 3-digit ID on card back — type it if the camera misses
- Cards are re-printable paper. No real instruments needed.

Designed for tabletop cameras, not phones.

Q&A: Anti-Memorization Design



Assessment

cards: neutral front — no name, no hint

- Pre-test and post-test use **matched parallel variants**
- Assessment cards show only the **AprilTag on the front**
- Pre and post use **different photo views** (A / B)
- Random seed ensures **unique layouts per session**

You cannot pass by memorizing card positions.

Q&A: Customizable Tray Module

```
{
  "instrument_id": "hospital_kelly_6in",
  "display_name": "Clamp Kelly 6\"",
  "family": "Clamping/occluding",
  "aliases": ["KELLY 6\"",
              "Kelly clamp 6"],
  "lookalike_pairs": [
    { "expected": "kelly_6in",
      "selected": "crile_5_5",
      "feedback": "Check serration
                  pattern..." }
  ],
  "required_items": [
    { "instrument": "kelly_6in",
      "quantity": 2 }
  ]
}
```

- Single editable JSON file per hospital tray
- Instruments, local names, aliases, counts
- Distractors, lookalike pairs, features
- No code changes — edit file, reprint cards
- One module = **2–4 hours** SPD educator time
- Reusable for every new hire, competency check

Q&A: Per-Hospital Data Burden

- Each hospital uses **different manufacturers, makes, and models**
- SPD lighting varies by **workstation, time of day, fixture condition**
- Tray layouts differ by **procedure type and surgeon preference**
- Cross-manufacturer validation needed — accuracy drops across brands
- Conservative estimate: **thousands of annotation hours per hospital**
- For **~5,500** US hospitals: industry-scale infrastructure problem

Not a one-project problem.

Q&A: Cost & Next Steps

Prototype Bill of Materials

Item	Cost
Printed cards (15–16 instruments)	\$15–30
Tray surface / mat	\$5–10
Webcam (if laptop cam insufficient)	\$10–15
Total	\$30–55

Assumes existing laptop or tablet.

Priorities for the Next Team

- 1 Run novice baseline study
- 2 Expert review of tray content
- 3 7-day retention check
- 4 SPD workplace pilot
- 5 Hospital partner for real module

\$53 in materials. Clinical validation first.

TrayGuard

Thank you. Questions?

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